

# HUMIDIFICATION

Chemical Engineering Lab 1I

02/04/2020

Group 2, Tuesday, PM section

Ahmad Abusubaih  
Fatimah Al Ibrahim  
Ahmed Alghamdi  
Velia Palma  
Christian Sanchez

# INTRODUCTION :

## The purpose of the experiment :

- **Obtaining the experimental coefficients of heat and mass transfer to study mass and heat transport in wetted wall column with using different flowrates and temperatures.**
- **Compare the obtained experimental values with the calculated values using theoretical equation.**

# THEORY:

$$q = h \cdot \Delta T \quad (1)$$

$$NA = kC \cdot \Delta cA \quad (2)$$

Where:

- **q** and **NA** are the heat and mass fluxes, respectively (energy or moles/area-time)
- **ΔT** is a characteristic temperature difference
- **ΔcA** is a characteristic concentration difference
- **h** and **kC** are the heat and mass transfer coefficients.

# CONT. THEORY:

**Mass transfer coefficient inside a circular pipe**

$$Sh = 0.023 \cdot Re^{0.83} \cdot Sc^{0.44} \quad (3)$$

$$Sh = kC \cdot L/DAB \quad (4)$$

Where:

- **Re** is the Reynolds number based on the pipe diameter
- **Sc** is the Schmidt number
- **Sh** is the Sherwood number
- **kC** is the mass transfer coefficient for a concentration driving force
- **L** is a characteristic length scale defined for the system under study
- **DAB** is the diffusion coefficient for the fluid through which the mass transport occurs

# CONT. THEORY:

$$(Gy) = k_c c (y_i - y) dA \quad (5)$$

Where:

- **d(Gy)** is the differential change in the moles of water in the gas over the differential column height
- **y** and **y<sub>i</sub>** are the mole fraction of water at the interface and in the gas
- **G** is the total gas flowrate
- **c** is the molar concentration of the gas
- **dA** is the area of differential column height

# CONT. THEORY:

## HEAT TRANSFER

$$d(GC_pT) = h (T_i - T)dA \quad (6)$$

Where:

- **C<sub>p</sub>** is the molar heat capacity of gas
- **T** is the temperature of gas
- **T<sub>i</sub>** is the temperature at the interface

# Apparatus:

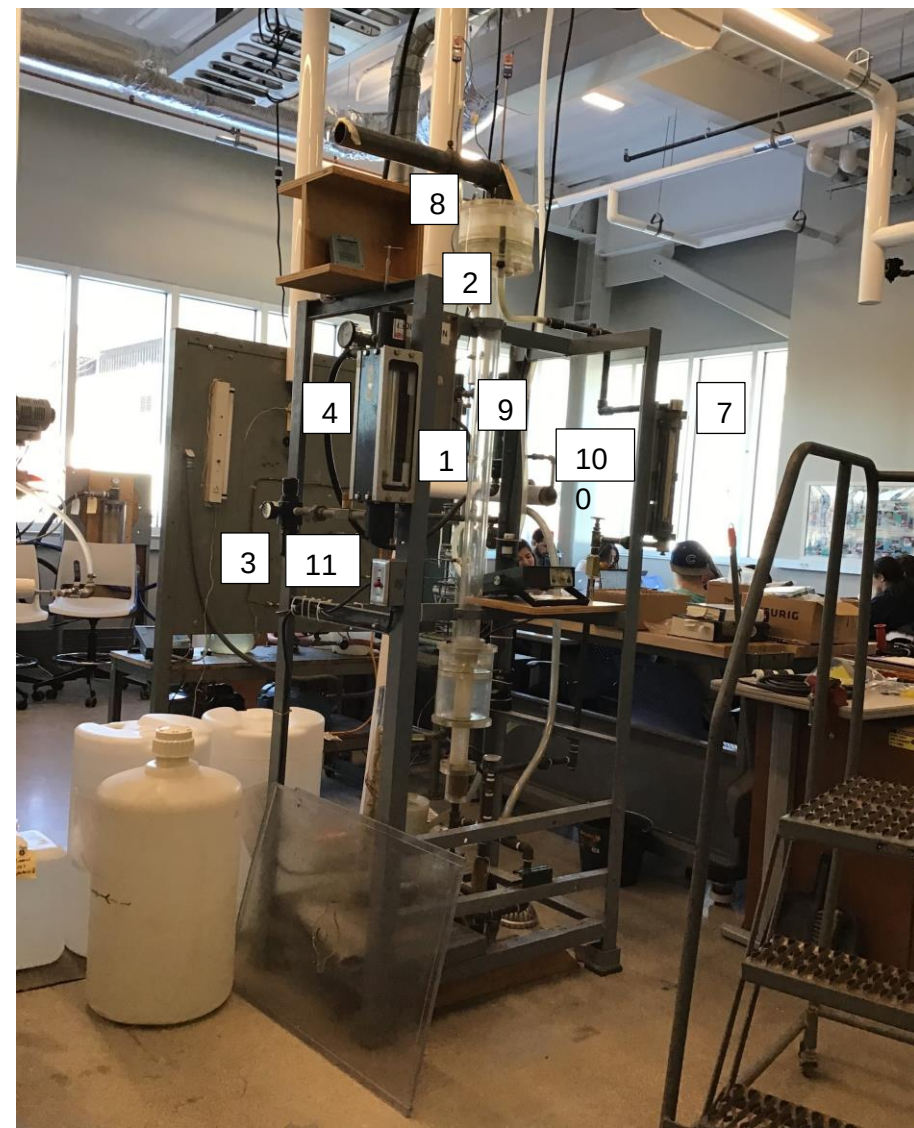


Figure 1: The important equipment of the experiment

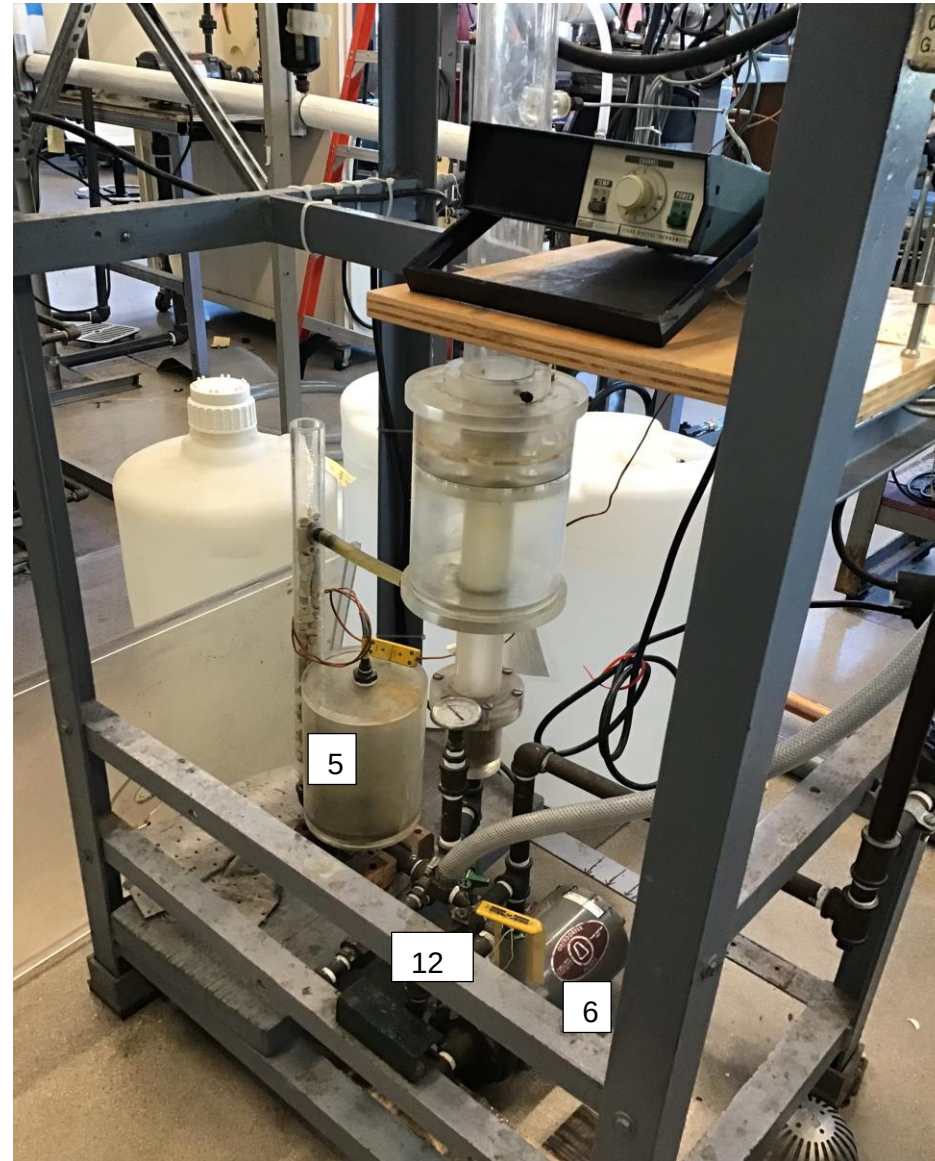


Figure 2: The important equipment of the experiment

## Cont.Apparatus:

**Table 1: The important components used in the experiment**

| #  | The important components              | Description                                                                                      |
|----|---------------------------------------|--------------------------------------------------------------------------------------------------|
| 1  | The wetted-wall humidification column | 2-inch diameter Plexiglas column 4 feet in length connected to an air flow and water feed system |
| 2  | House Supply                          | Supplies the system with air                                                                     |
| 3  | Pressure regulator                    | Manipulates the pressure of air                                                                  |
| 4  | Air rotameter                         | Measures the flowrate of air (unit and mx                                                        |
| 5  | Tank                                  | Source of water supplied to the system                                                           |
| 6  | Centrifugal pump                      | Pumps the water from the holding tank into the system                                            |
| 7  | Water rotameter                       | Measures the flowrate of water (unit and mx                                                      |
| 8  | Weir                                  | Distributes the water to the side of the top of the column                                       |
| 9  | Heat Exchanger                        | Heats incoming air                                                                               |
| 10 | Valve 1                               | To control the water flow rate                                                                   |
| 11 | Valve 2                               | To control the air flow rate                                                                     |
| 12 | Valve 3                               | Drain valve                                                                                      |
| 13 | Steam Valve                           | To increase the temperature of air                                                               |



# MATERIAL & SUPPLIES :

**Table 2: The important materials and supplies used in the experiment**

| # | The Material and Supplies | Description                                                                                                |
|---|---------------------------|------------------------------------------------------------------------------------------------------------|
| 1 | Compressed Air            |                                                                                                            |
| 2 | Water                     |                                                                                                            |
| 3 | Electronic hydrometer     | Measures the temperature of the air exiting the humidification column                                      |
| 4 | Thermocouple              | Measures the temperature of incoming air and water, and the temperature of the water throughout the column |
| 5 | Ladder                    | Used to reach the point where humidity of incoming air measured                                            |
| 6 | Steam                     | the shell-side to heat the air                                                                             |

# PROCEDURE

## Before starting the experiment:

- Measure the humidity of the incoming air using the electronic hygrometer
- Measure the temperature of the incoming and outgoing air

## The experiment:

- Fill the tank with water
- Turn on water flow at a sufficiently high flow rate
- Turn on the house supply in order to establish the unheated air flow at **3 SCFPM**
- Allow the temperature of the water to reach the steady state
- Record the water and air temperatures at the inlet, along the column and the air humidity at the outlet
- Repeat the same steps using two different flow rates(**6 and 9 SCFPM**)
- Set the air flow to **6 SCFPM**
- turn on the steam in order to increase the temperature of the air flow
- Repeat step 3 through 6 using the heated air flow
- Turn off air and water flow
- Drain water and clean up

# EXPERIMENTAL CHALLENGES



- Obtaining accurate humidity of incoming air
- Perfecting of the weird placement
- Small pump leak

# SAFETY:



- Make sure to wear safety glasses
- Ask for mask if the smell of the boiler is too strong.
- Careful with hot pipes from the steam line.
- Ask for ear protection if the sound of the boiler is too loud.
- Keep an eye for any leakage.
- Make sure to not exceed the maximum flow rate of air and water.
- Do not overfill water tank



**THANK YOU.**

FEEL FREE TO ASK QUESTIONS

